

GoFarther AI — File Operations

Complete reference: 24 operations, 15 file types, full prompt + backend spec

1. All 24 Operations

Modification Operations (15)

Operation	Description	Output	Files
edit	Modify content: add/delete rows, change values, add formulas	File	Single
chart	Create visualization: bar, pie, line, scatter, histogram	PNG	Single
convert	Change format: xlsx→pdf, csv→xlsx, etc.	File	Single
merge	Combine multiple files into one	File	Multi (file_ids)
filter	Extract rows matching criteria	File	Single
compare	Diff two files: added, removed, changed	PDF	Multi (file_ids)
reconcile	Match transactions between 2 sources	XLSX (styled)	Multi (file_ids)
clean	Remove duplicates, fix formatting, standardize	File	Single
split	Split one file into multiple sheets/groups	XLSX	Single
rename	Rename/reorder columns or sheets	File	Single
ocr	Extract text from images/scanned PDFs via Vision	Text (chat)	Single
append	Add rows WITHOUT touching existing data	File	Single
pivot	Create pivot table with aggregation + formulas	XLSX	Single
translate	Translate entire file preserving structure	File	Single
highlight	Conditional formatting with color-coded cells	XLSX (styled)	Single

Read Operations (6) — returns text in chat, no file download

Operation	Description	Output	Files
summarize	Plain English summary, key insight first	Text (chat)	Single
analyze	Deep analysis: trends, patterns, outliers	Text (chat)	Single
find	Locate specific data, return row/column/section	Text (chat)	Single
extract	Pull specific data points only	Text (chat)	Single
answer	Answer a question about the file	Text (chat)	Single
validate	Data quality audit: errors, dupes, inconsistencies	Text (chat)	Single

Meta Operations (3)

Operation	Description	Output	Files
batch	Run same operation on multiple files, returns ZIP	ZIP	Multi (file_ids)
chain	Run multiple ops sequentially on one file	File	Single
create	Generate new file from description	File	N/A

2. Supported File Types (15)

Format	Category	Support	Library
xlsx, xls	Spreadsheets	Read + Write + Formulas	openpyxl
csv, tsv	Delimited data	Read + Write	Native

Format	Category	Support	Library
pdf	Documents	Read + Write	PyMuPDF + reportlab
docx	Word documents	Read + Write	python-docx
txt	Plain text	Read + Write	Native
pptx	Presentations	Read + Write	python-pptx
json	Structured data	Read + Write	Native JSON
html	Web pages	Read + Write	Regex strip
md	Markdown	Read + Write	Passthrough
rtf	Rich text	Read + Write	RTF strip/wrap
zip	Archives	Read only	zipfile
png, jpg, jpeg	Images	Read/OCR	Claude Vision
webp	Modern images	Read/OCR	Claude Vision
heic, heif	iPhone photos	Read/OCR	Pillow → PNG → Vision

3. Action JSON Formats

Create:

```
{"type": "create_file", "target": "description", "text": "pdf|xlsx|docx|csv|txt|pptx"}
```

Modify (single file):

```
{"type": "modify_file", "target": "<operation>", "text": "instructions"}
```

Modify (multi-file — merge/compare/reconcile):

```
{"type": "modify_file", "target": "merge", "file_ids": ["file1.xlsx", "file2.xlsx"], "text": "instructions"}
```

file_ids uses FILENAMES — frontend resolves to internal IDs via registry.

Read:

```
{"type": "modify_file", "target": "summarize|analyze|find|extract|answer|ocr|validate", "text": "question"}
```

Batch:

```
{"type": "modify_file", "target": "batch", "text": "convert:", "key": "pdf", "file_ids": ["a.xlsx", "b.xlsx"]}
```

Chain:

```
{"type": "modify_file", "target": "chain", "chain_ops": [{"operation": "clean"}, {"operation": "filter", "instructions": "..."}, {"operation": "convert", "target_format": "pdf"}]}
```

Supported chain steps: edit, clean, filter, convert, rename, chart, split, append, pivot, translate, highlight

4. Architecture

Component	Details
Frontend (types.ts)	parseAction() extracts JSON from AI response. ActionPayload includes file_ids, chain_ops, save_contact.
Frontend (useChat.ts)	File registry maps filenames→IDs. Resolves multi-file ops. Auto-uploads user attachments. Validates chain_ops. Shows progress.
Frontend (ai.ts)	createFile(), uploadFile(), modifyFile() — async job pattern with 3s polling, progress callback, 3min timeout.
Backend (ghost_tools.py)	POST /create-file-async, /upload-file, /modify-file-async, GET /job-status/{id}, /download/{id}. All async via JOB_STORE.
File Storage	In-memory cache (500 files) + Cloudflare R2. Ownership via owner_email. LRU eviction.
AI Layer	_ask_claude() with 2x retry + exponential backoff. _validate_and_fix_spreadsheet_json() for structure repair.

5. Infrastructure Features

Feature	Implementation
File Naming	_smart_filename(): description_YYYY-MM-DD.ext for creates, original_action_YYYY-MM-DD.ext for modifies. Fallback to 'file' if slug is
File Registry	Per-session Map<filename, file_id> persisted in AsyncStorage. 3-level resolution: exact → base name → partial match.
Auto-Upload	User file attachments auto-uploaded to backend when modify op is triggered. Shows 'Uploading file...' feedback.
Retry Logic	_ask_claude() retries 2x with 1s/2s backoff on timeout, HTTP error, or empty response.
JSON Validation	_validate_and_fix_spreadsheet_json() checks headers/rows structure. Asks Claude to fix if wrong. Handles {data:[...]} → [{...}] normaliz
Progress	JOB_STORE[id]['progress'] updated during OCR (per batch), chain (per step), batch (per file). Frontend polls and displays.
Batch ZIP	All batch output files zipped into single download. Individual files also stored for per-file access.
Chain Reporting	completed_steps[] + failed_step in job response. Frontend shows 'clean → filter → convert' with failure indicator.
Security	Owner-email check on all file access. Sensitive data masking in chat. Password-protected file handling.
Highlight Styling	Shared _build_highlighted_xlsx() used in both standalone and chain. Color map: red/green/yellow/blue/orange.

6. Quality Rules (from prompt)

Type	Requirements
xlsx	Real formulas (SUM, AVERAGE, IF, VLOOKUP), named sheets, freeze header row, formatted columns
pdf	Proper headings, sections, page numbers, professional typography
docx	Heading styles (H1/H2/H3), proper margins, professional structure
pptx	Title slide + content slides + closing, minimum 5 slides unless specified
csv	Header row, clean delimiters, no formulas
Financial	Always formula-driven, never static values, must balance/total correctly

7. Backend Endpoints

Method	Path	Description
POST	/api/ghost/tools/create-file-async	Create new file (pdf/xlsx/docx/csv/txt/pptx)
POST	/api/ghost/tools/upload-file	Upload user file to FILE_STORE
POST	/api/ghost/tools/modify-file-async	Run any of 24 operations on file(s)
GET	/api/ghost/tools/job-status/{id}	Poll job progress and result
GET	/api/ghost/tools/download/{id}	Download file by ID (ownership checked)

18 commits · 24 operations · 15 file types · 10 output formats · 11 chain steps